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Design a differential amplifier operating on a lower supply voltage of 1.8 V, utilizing a wide-swing (high-compliance) cascode current mirror load. Analyze the trade-offs made to maximize the output voltage swing while maintaining a minimum voltage gain of 40 dB

ANALOG IC Design-Mini Project

MANISH

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ABSTRACT

This Mini-project presents the design of a low power 1.8 V fully differential amplifier in the SkyWater 130 nm process, utilizing a PMOS input differential pair and an NMOS wide swing cascode active load.The target was a minimum differential gain of 40 dB while keeping all transistors in saturation and maximizing output swing. Also validated across process corners, temperatures (-40 °C to 125 °C), and Supply variations using ngspice, the design achieves robust stability and reliable current mirroring while highlighting key trade-offs between gain and cascode headroom.



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OBJECTIVES

- Design a differential amplifier operating on a lower supply voltage of 1.8 V.
- With a wide-swing cascode current mirror as load.
- Design the whole circuit in a such way that have a minimum voltage gain of 40 dB.
- Analyzing the trade-of made to maximize the output voltage swing.
- Maintaining all Transistor in saturation mode.

WORKING

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- **Differential Current Steering**:

When the input voltage on one branch rises relative to the other, the circuit doesn't create new current it simply balance the current into one branch and away from the other. So like if left PMOS gets more current (I_tail / 2 + Δ i), the right PMOS automatically gets less (I_tail / 2 - Δi) .

- **Active Load Current Mirroring**:

The right output node has two opposing forces:

The top PMOS branch trying to push its current (i_d2) into the output.

The bottom NMOS load pulling the mirrored current (i_d1) down to ground.

Because the two currents move in opposite directions (+Δ i vs -Δi), they push and pull against the node's high output resistance, converting current difference into a huge single-ended voltage swing

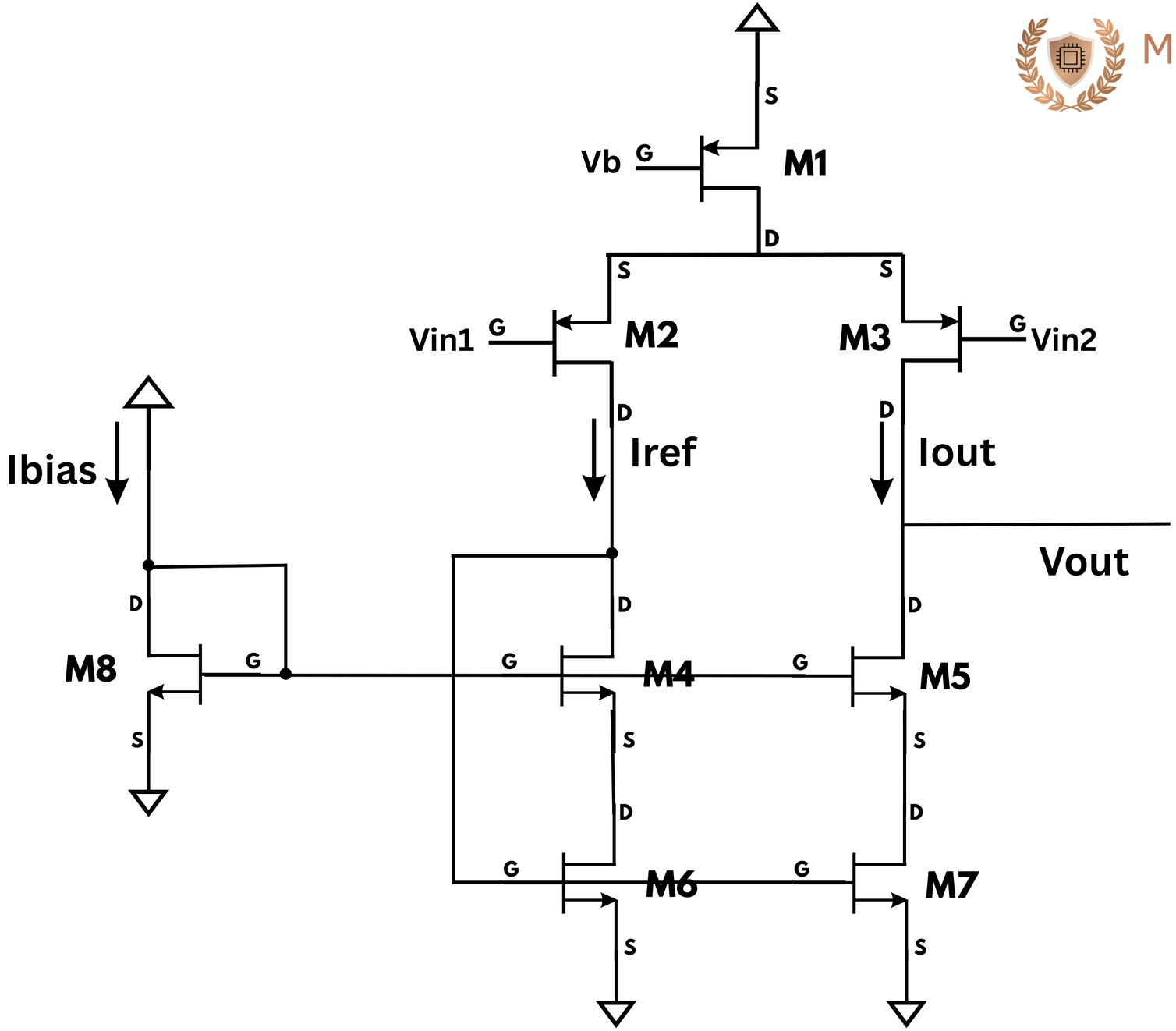
- **Cascode Impedance Shielding**:

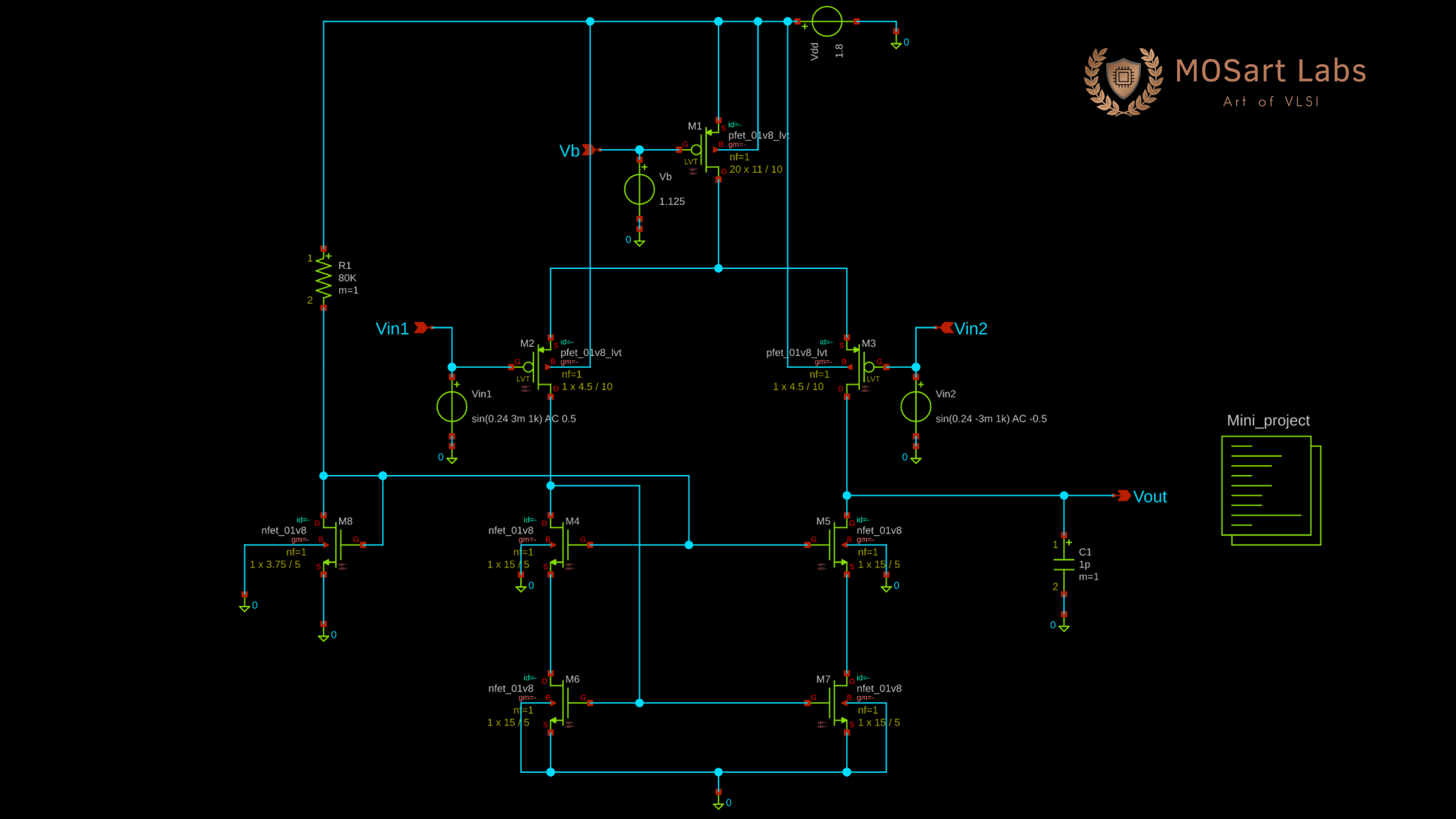
A cascode transistor is stacked directly on top of the bottom mirror transistor. It acts as an impedance multiplier and voltage buffer. It isolates the bottom transistor's drain, holding that node virtually constant regardless of how wildly Vout fluctuates.Because the bottom transistor sees a stable voltage, its current is constant.

- **Headroom Preservation**:

Wide-swing gate bias holds the bottom mirror drain-source voltage at exactly , preventing premature saturation loss. and makes Vds= Vov.

DESIGN


S
Vb
G
M1
D
S
S
Vin1
G
M2
M3
G
Vin2
D
D
Ibias
Iref
Iout
Vout
D
D
D
M8
G
G
G
M4
M5
S
S
S
D
D
G
G
M6
M7
S
S


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Vdd
1.8
0
M1
pfet_01v8_lvt
id=-
gm=-
nf=1
20 x 11 / 10
LVT
Vb
Vb
1.125
R1
80K
m=1
Vin1
M2
pfet_01v8_lvt
nf=1
1 x 4.5 / 10
Vin1
sin(0.24 3m 1k) AC 0.5
M3
pfet_01v8_lvt
nf=1
1 x 4.5 / 10
Vin2
Vin2
sin(0.24 -3m 1k) AC -0.5
Vout
M8
nfet_01v8
nf=1
1 x 3.75 / 5
M4
nfet_01v8
nf=1
1 x 15 / 5
M5
nfet_01v8
nf=1
1 x 15 / 5
M6
nfet_01v8
nf=1
1 x 15 / 5
M7
nfet_01v8
nf=1
1 x 15 / 5
C1
1p
m=1
Mini_project

DESIGN

Assumed Vov=0.2V, MnCox = 165µ A/V², Vthn = 0.53V, Vthp = 0.43V, MpCox = 45µ A/V²

Iss=20µA and VDD = 1.8V, Av=40db

From the top Transistor M1 , the (W/L)1 is ,

$$\frac{2\,(Id)}{MpCox \times Vov^2} = \frac{2\left(20 \times 10^{-6}\right)}{45 \times 10^{-6} \times (0.2)^2} = \boxed{22.22 \approx 22}$$

Vb is ,

$$Vsg = Vov + Vthp \qquad Vs - Vg = Vsg$$

$$Vg = Vb = Vs - (Vov + Vthp) \qquad Vb = 1.8 - 0.2 - 0.43 \qquad \boxed{Vb = 1.17V}$$

For the main current mirror M4 & M5 , and the cascade mirror M6 & M7 the (W/L) is ,

n = 1

$$\frac{2\,(Id)}{MnCox \times Vov^2} = \frac{2\left(10 \times 10^{-6}\right)}{165 \times 10^{-6} \times (0.2)^2} = \boxed{3.03 \approx 3}$$

For the Diode connected Transistor M8 , the (W/L)8 is ,

$$\frac{(\frac{W}{L})}{(n+1)^2} = \frac{(3)}{(1+1)^2} = \boxed{0.75}$$

For the Output Voltage ,

$$Vout, min = 2Vov \qquad = 2 \times 0.2 \qquad Vout, min = \boxed{0.4V}$$

$$Vout, max = VDD - 2Vov \qquad = 1.8 - 0.4 \qquad Vout, max = \boxed{1.4V}$$

$$Vout, swing = Vout, max - Vout, min$$

$$Vout, swing = 1.4V - 0.4V$$

$$Vout, swing = \boxed{1V}$$

For the Input Voltage Vin,

$$Vin = VDD - 2Vov - Vthp \qquad = 1.8 - 0.4 - 0.43$$

$$Vin = \boxed{0.97V}$$

Form the Voltage gain ,

$$Av = \frac{Vout}{Vin, diff} = -Gm.\,Rout$$

$Gm = gm, in = gm, diff$ (Transconductance of input diff. pair transistor)

$$Rout = Rup \| Rdown$$

$$Rup = ro, in = ro, diff$$

(In small - signal analysis gate and source terminal of M1 are AC ground
so vsg=0, only M2 & M3 will be considered for pull-up impedence)

$$Rdown = gm5ro5.ro7$$

$$Rdown = gm.ro^2$$

(For pull-down impedence its the cascade current mirror design)

$$Av = -Gm.Rout$$

$$Rup = ro3 = ro2 = \frac{1}{\lambda(Id)} = \frac{1}{0.02 \times 10 \times 10^{-6}} = \boxed{5M\Omega}$$

$$gm5 = gm4 = gm6 = gm7 = gm8 = gm, load = \frac{2(Id)}{Vov} = \frac{2 \times 10 \times 10^{-6}}{0.2} = \boxed{100 \times 10^{-6} S}$$

$$ro5 = ro4 = ro6 = ro7 = ro8 = \frac{1}{\lambda(Id)} = \frac{1}{0.02 \times 10 \times 10^{-6}} = \boxed{5M\Omega}$$

$$Rdown = gm.ro^2 = 100 \times 10^{-6} \times 5M \times 5M = \boxed{2.5G\Omega}$$

$$Rout = Rup\,||Rdown = 5M\,||2.5G = \boxed{4.99M \approx 5M\Omega}$$

$$Av = 40db = 10^{\frac{40}{20}} = \boxed{100}$$

$$Gm = gm2 = gm3 = \frac{Av}{Rout} = \frac{100}{5M} = \boxed{20 \times 10^{-6} S}$$

For W/L of differential pair Transistor,

$$\frac{W}{L} = \frac{gm3^2}{2 \times Id \times MpCox} = \frac{(20 \times 10^{-6})^2}{2 \times 10 \times 10^{-6} \times 45 \times 10^{-6}} = \frac{4}{9} = \boxed{0.444}$$

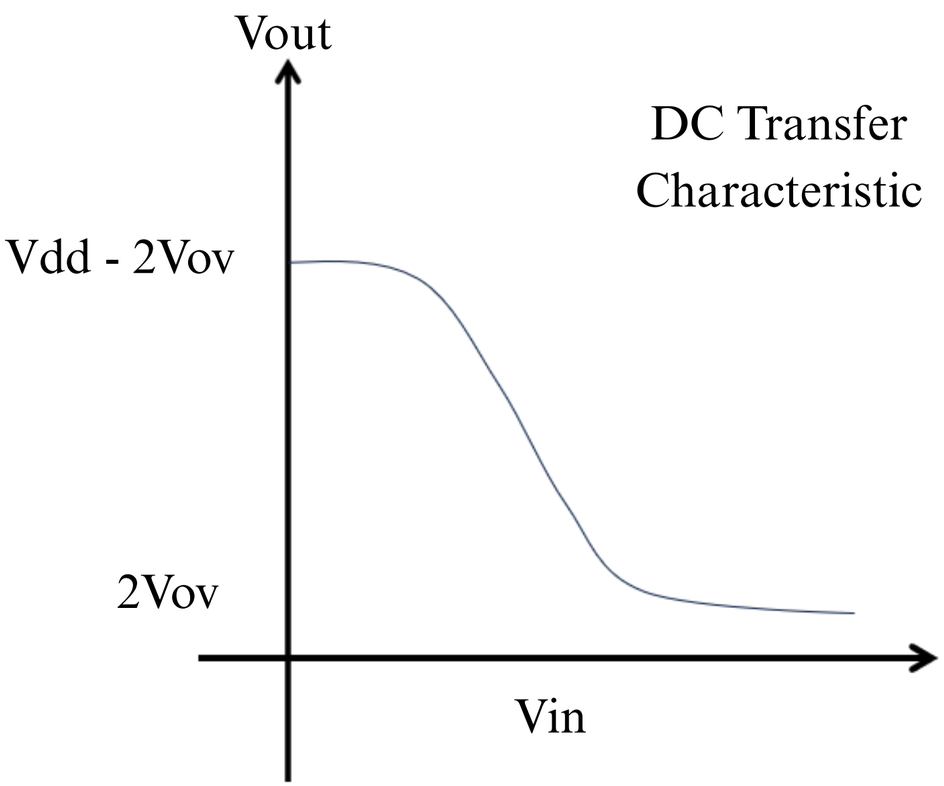
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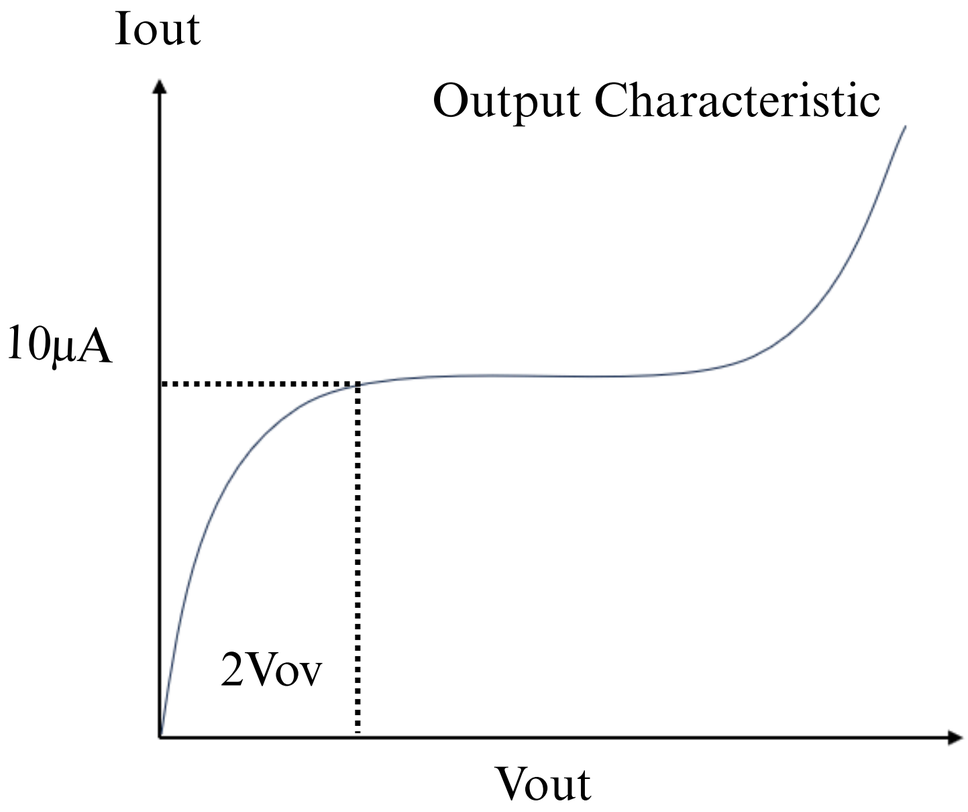
DESIGN SPECIFICATION

Transistor	Type	Id	Vov	W/L	W	L	ro
M1	PMOS	20µ	0.2V	22	20 X 11	10	2.5M
M2	PMOS	10µ	0.2V	0.45	4.5	10	5M
M3	PMOS	10µ	0.2V	0.45	4.5	10	5M
M4	NMOS	10µ	0.2V	3	15	5	5M
M5	NMOS	10µ	0.2V	3	15	5	5M
M6	NMOS	10µ	0.2V	3	15	5	5M
M7	NMOS	10µ	0.2V	3	15	5	5M
M8	NMOS	10µ	0.2V	0.75	3.75	5	5M

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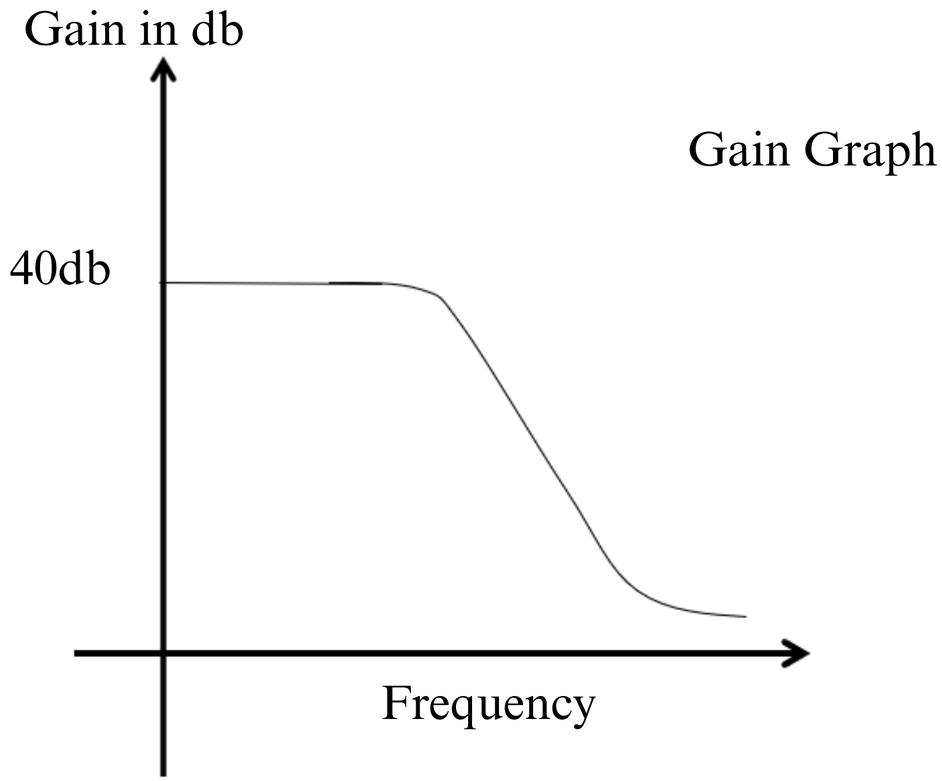
EXPECTED WAVEFORM





EXPECTED WAVEFORM

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EXPECTED PERFORMANCE

40db	1V	54µW	20µS
DC Voltage Gain	Output Swing (p-p)	Power Dissipation	Input Transistor Transconductance

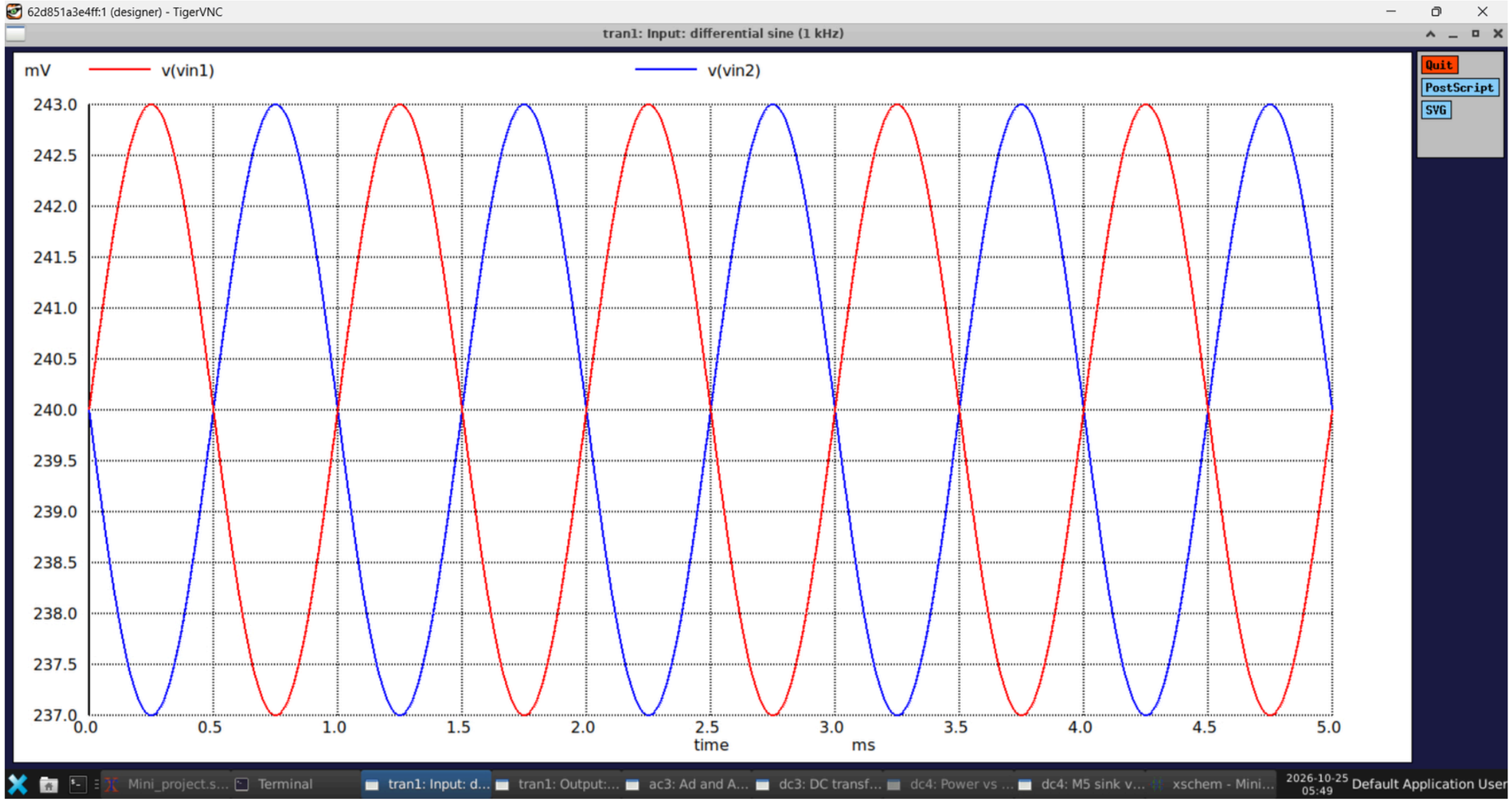
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OPERATING POINTS

Transistor	I (in µ)	gm	gds	Vds	Vov	Margin	Mode
M1	20µ	167µ	21.1µ	0.1875	0.18	7.5m	Edge of saturation
M2	10µ	26.3µ	0.301µ	0.9149	0.691	223.9m	Saturation
M3	10µ	26.3µ	0.301µ	0.9149	0.691	223.9m	Saturation
M4	10µ	116µ	0.381µ	0.5323	0.155	377.3m	Saturation
M5	10	116µ	0.381µ	0.5323	0.155	377.3m	Saturation
M6	10µ	107µ	11µ	0.165	0.156	9m	Edge of saturation
M7	10µ	107µ	11µ	0.165	0.156	9m	Edge of saturation
M8	11.2µ	59.79µ	0.24µ	0.89	0.3	590m	Saturation

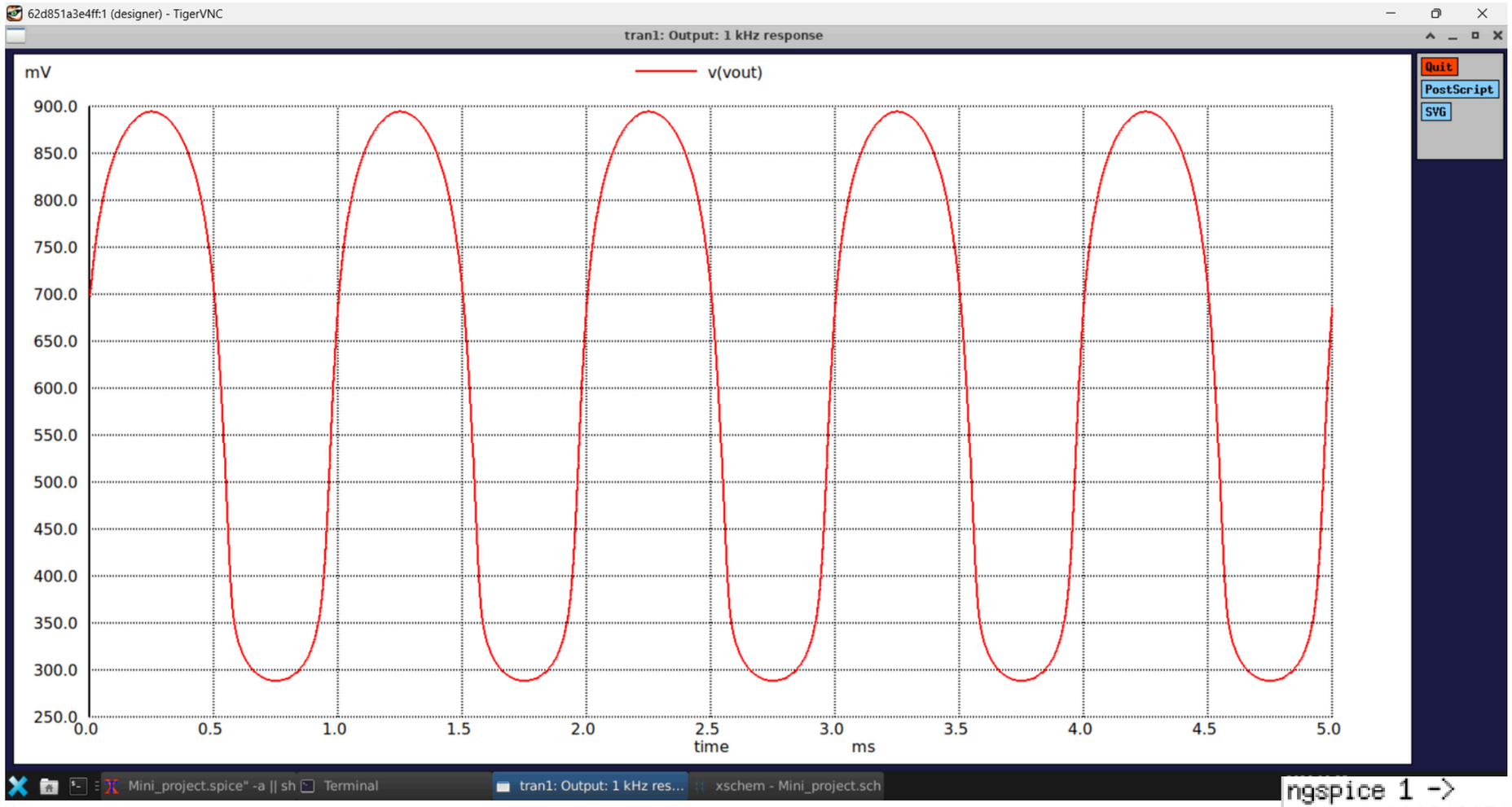
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SIMULATED WAVEFORM



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$$Vout, swing = (0.897917 - 0.290625) = 0.607292V$$

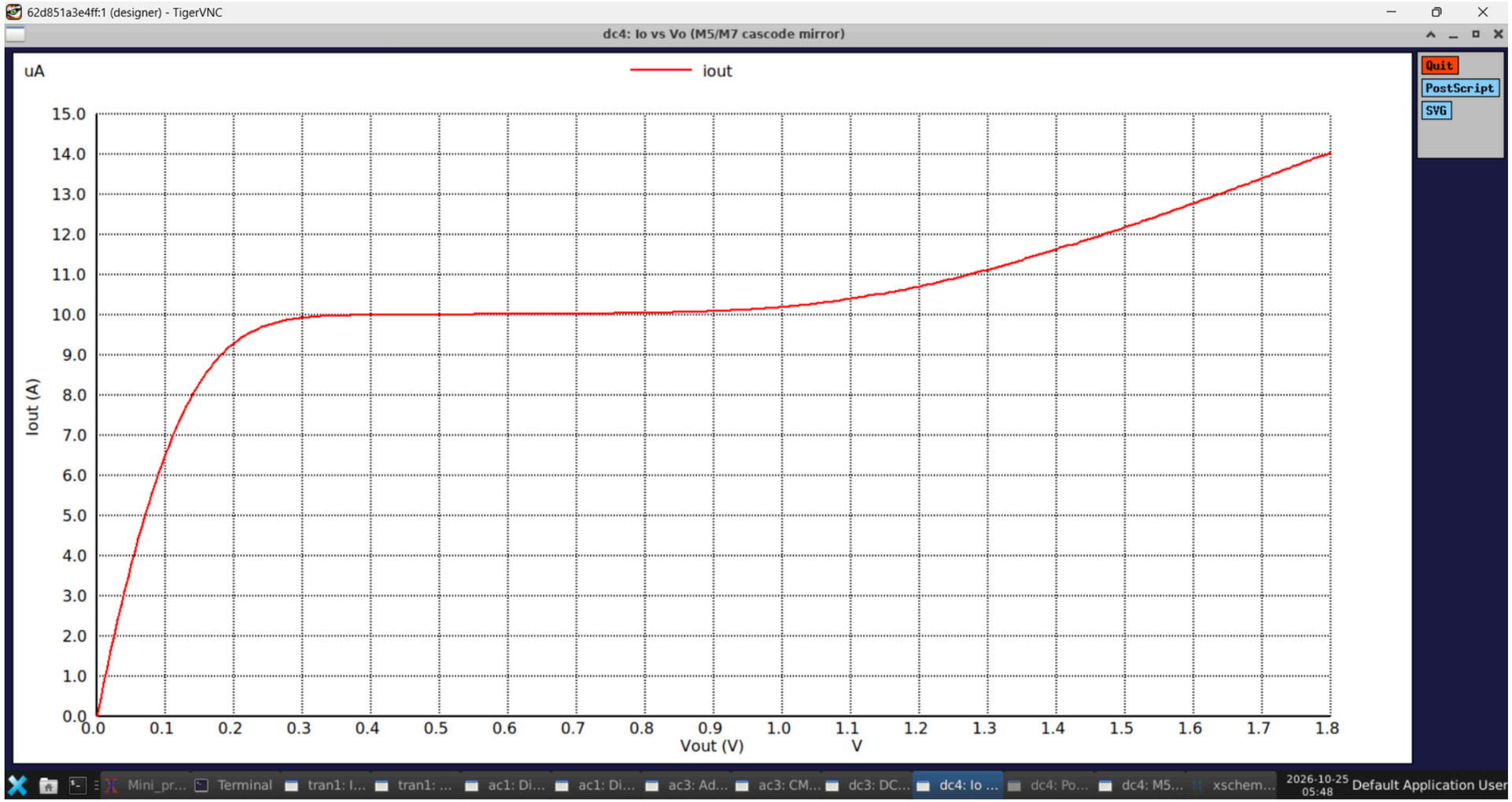



```
ngspice 1 ->
x0 = 0.000248031, y0 = 0.897917

x0 = 0.000748031, y0 = 0.290625
```

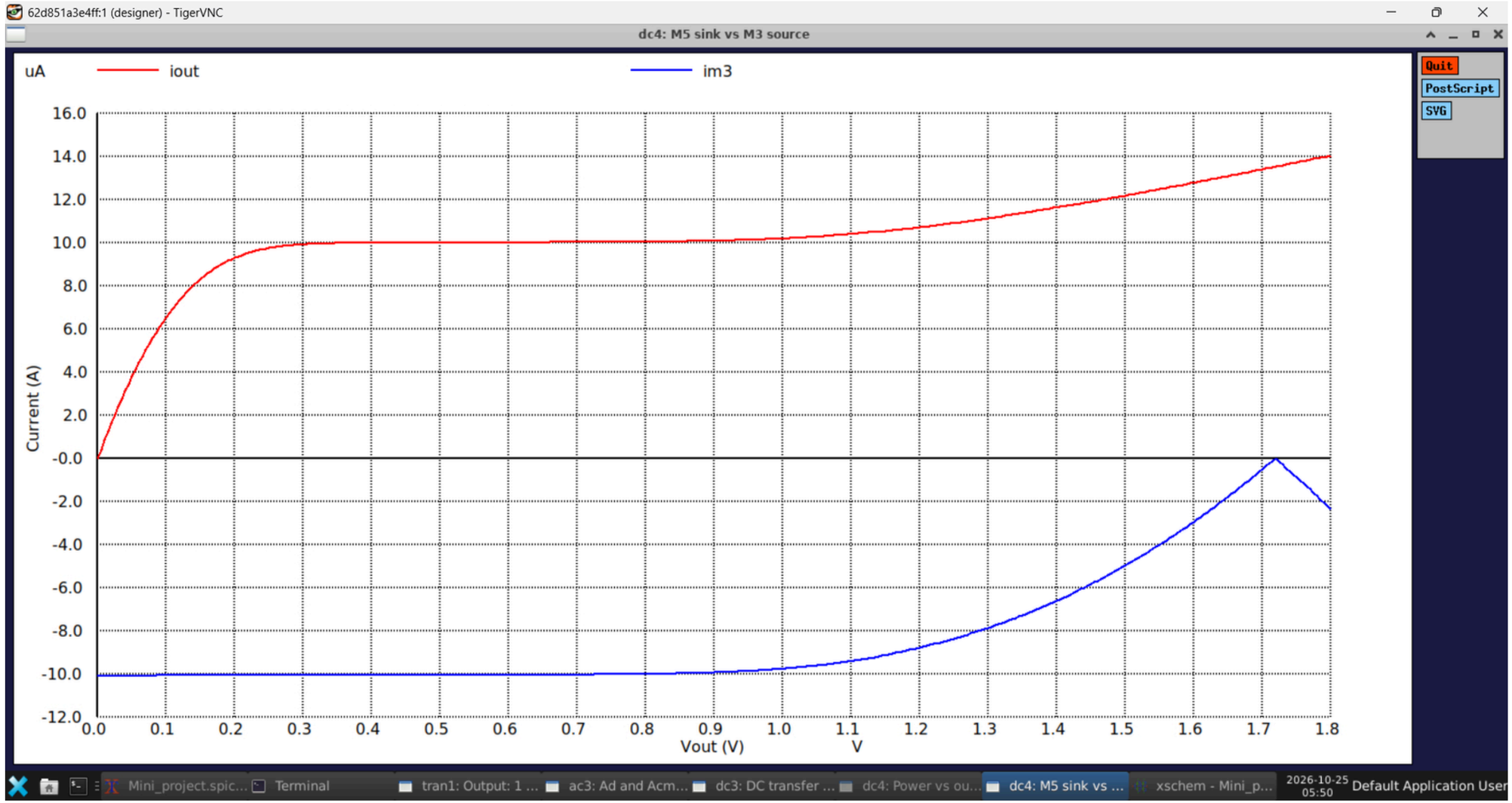
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Iout vs Vout



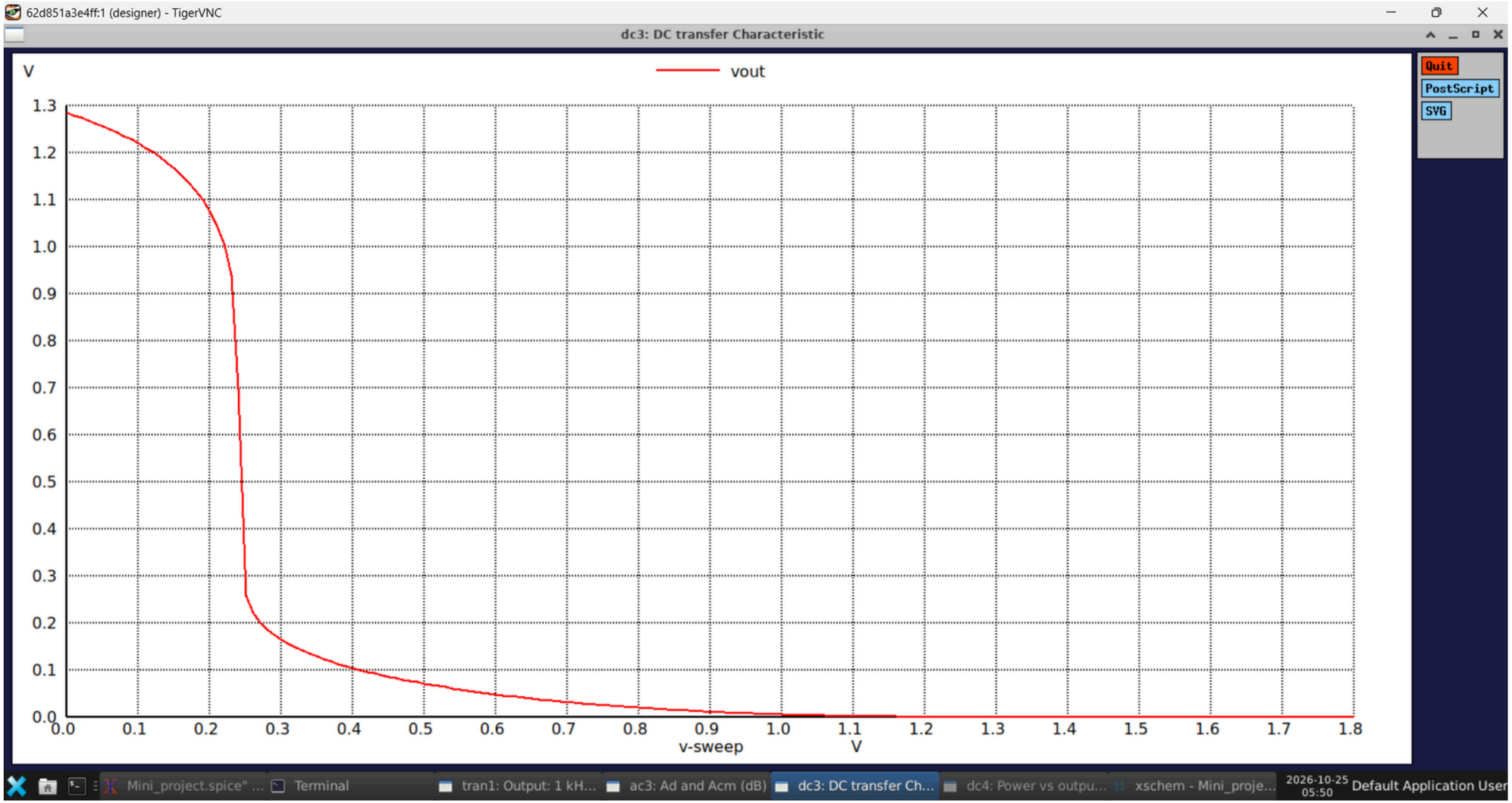
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Iout & Iref vs Vout



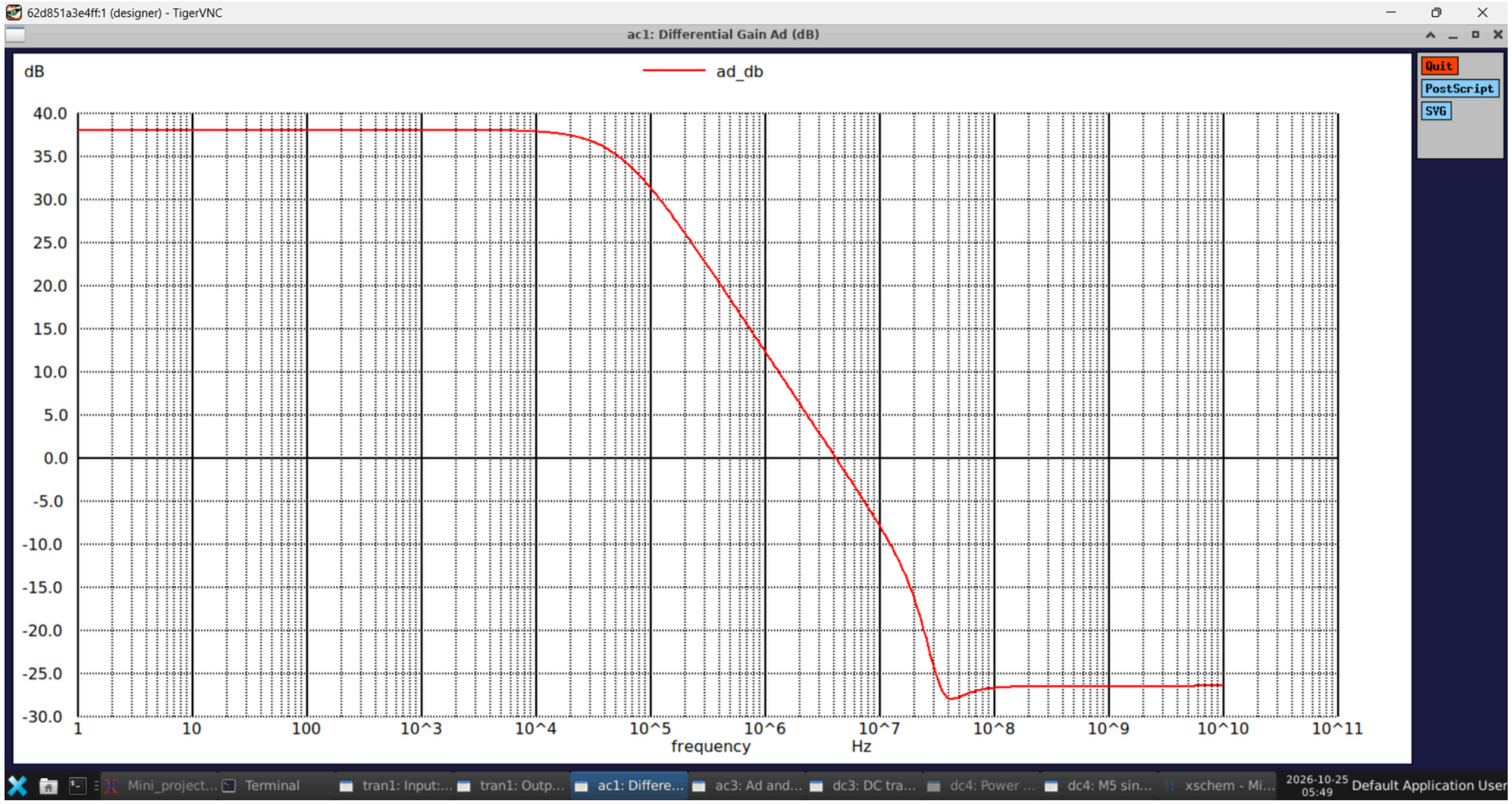
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DC characteristic



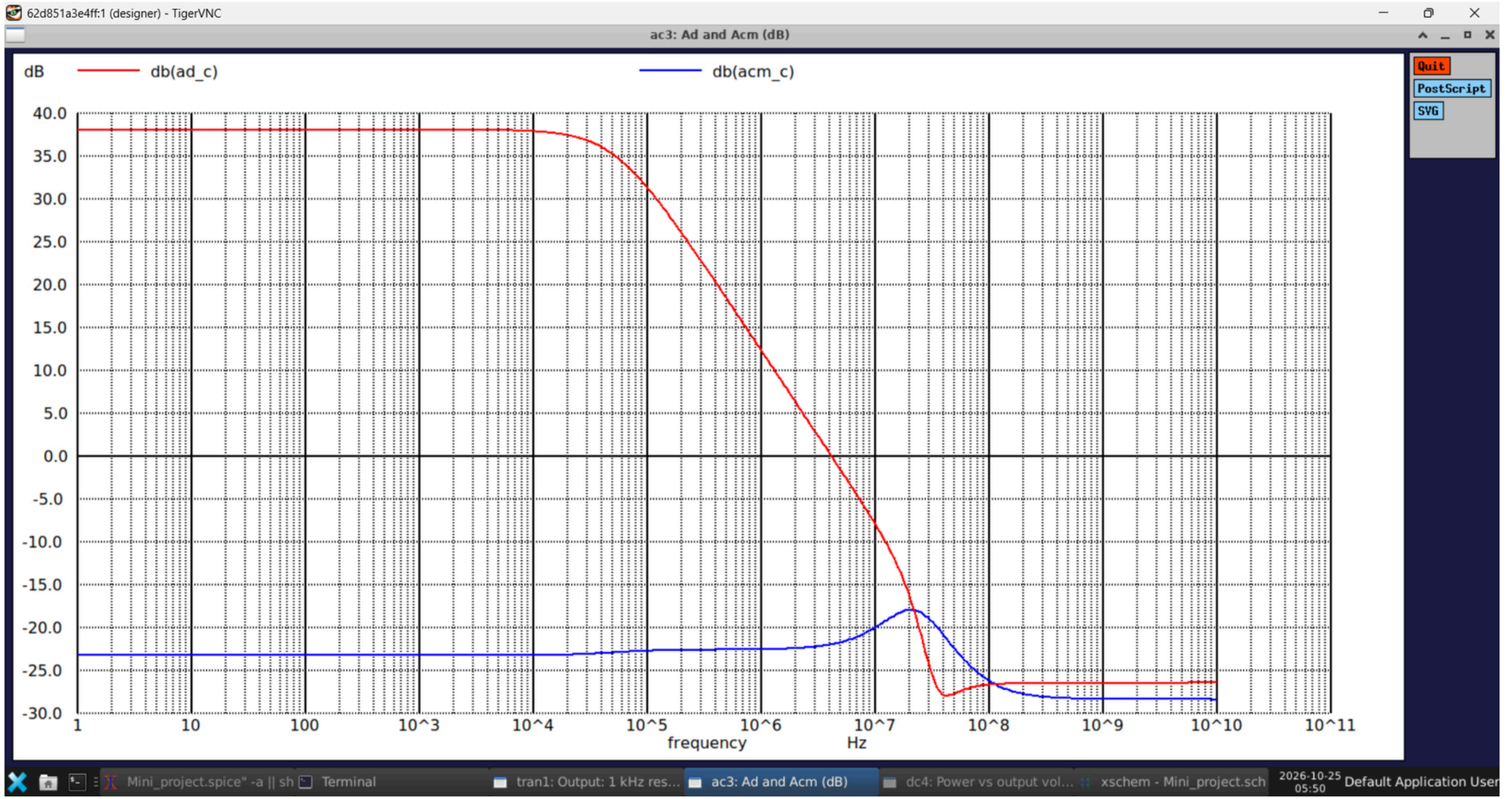
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Differential Gain



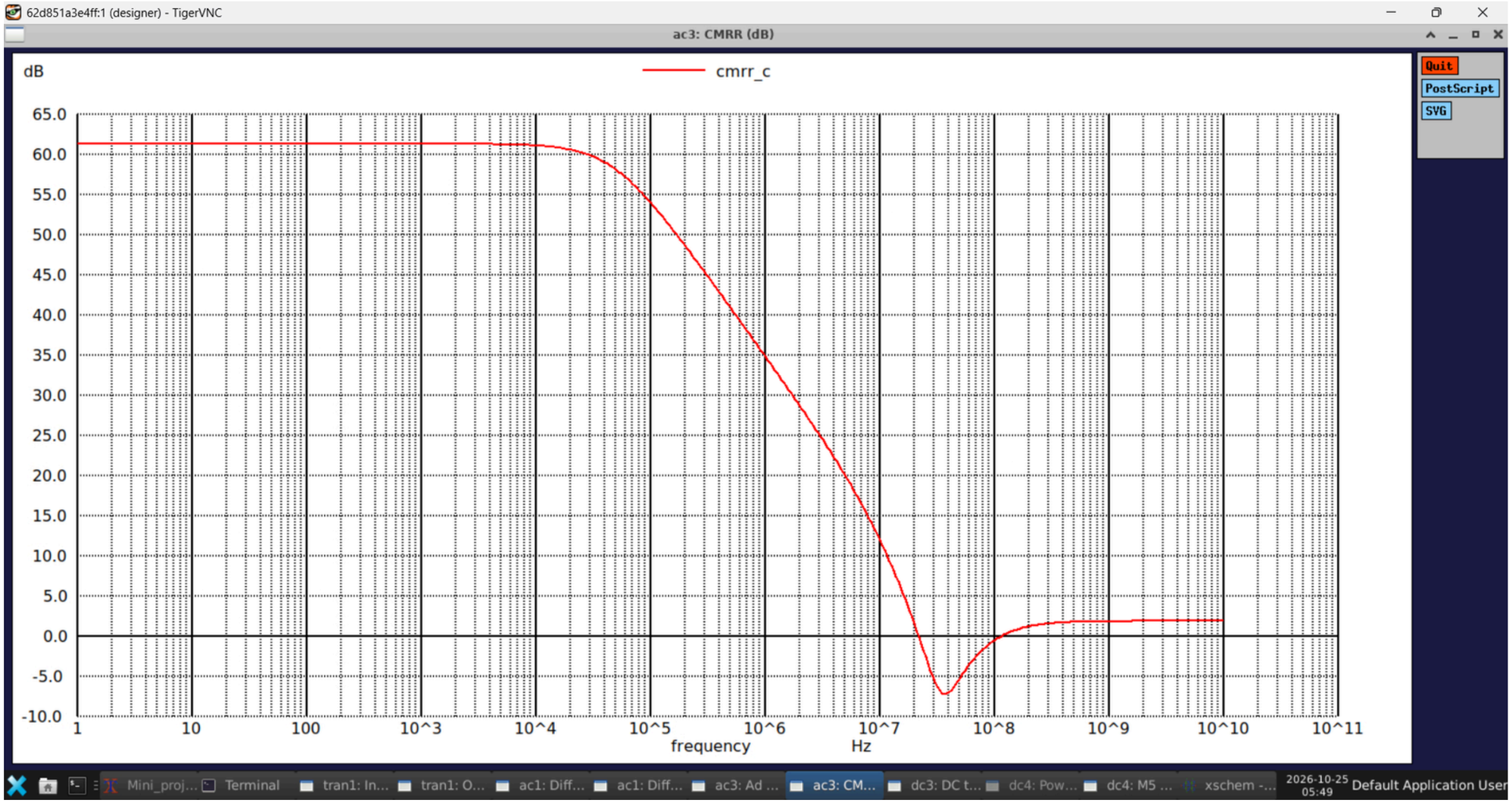
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Ad & Av,cm



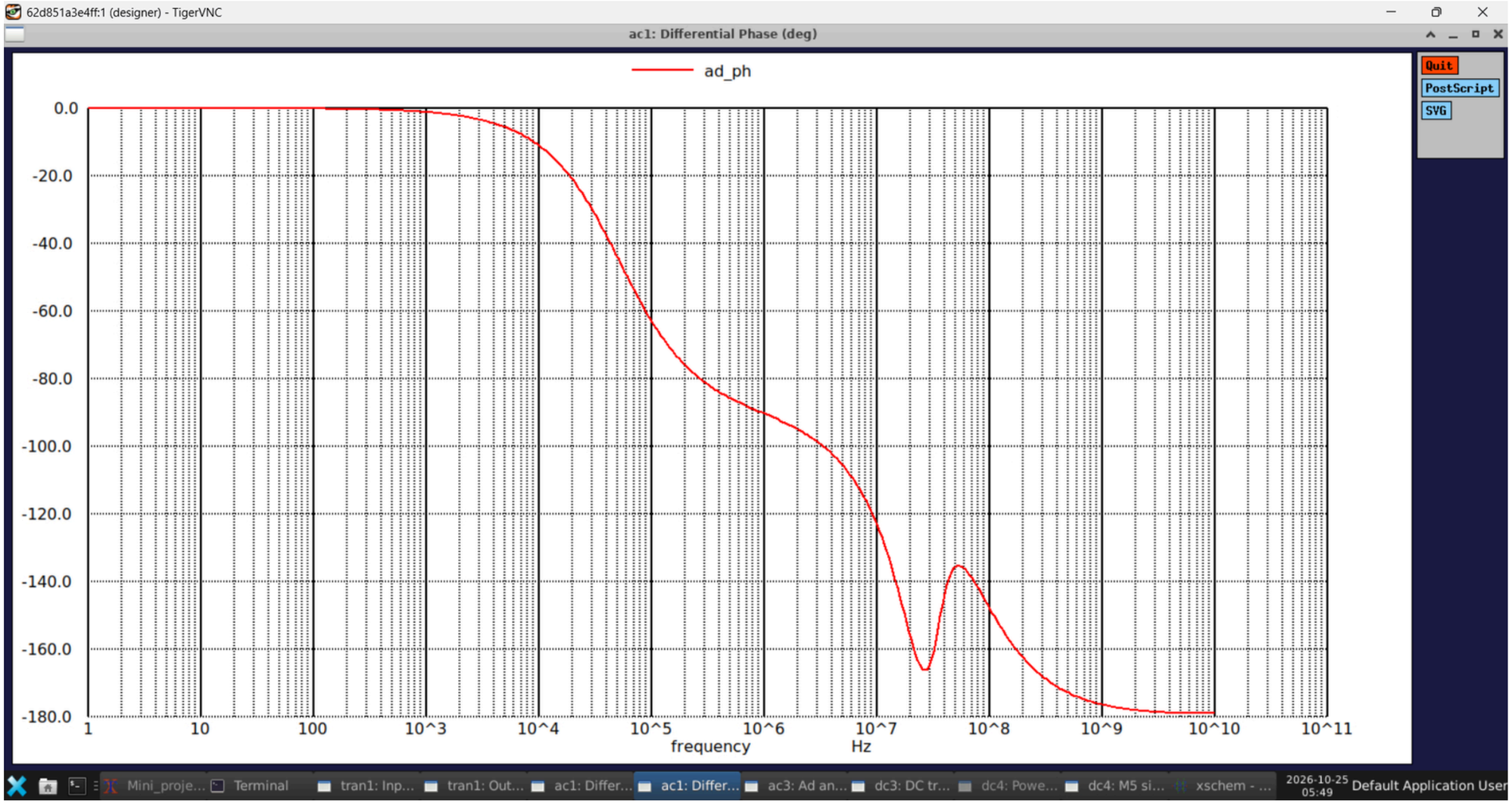
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CMRR



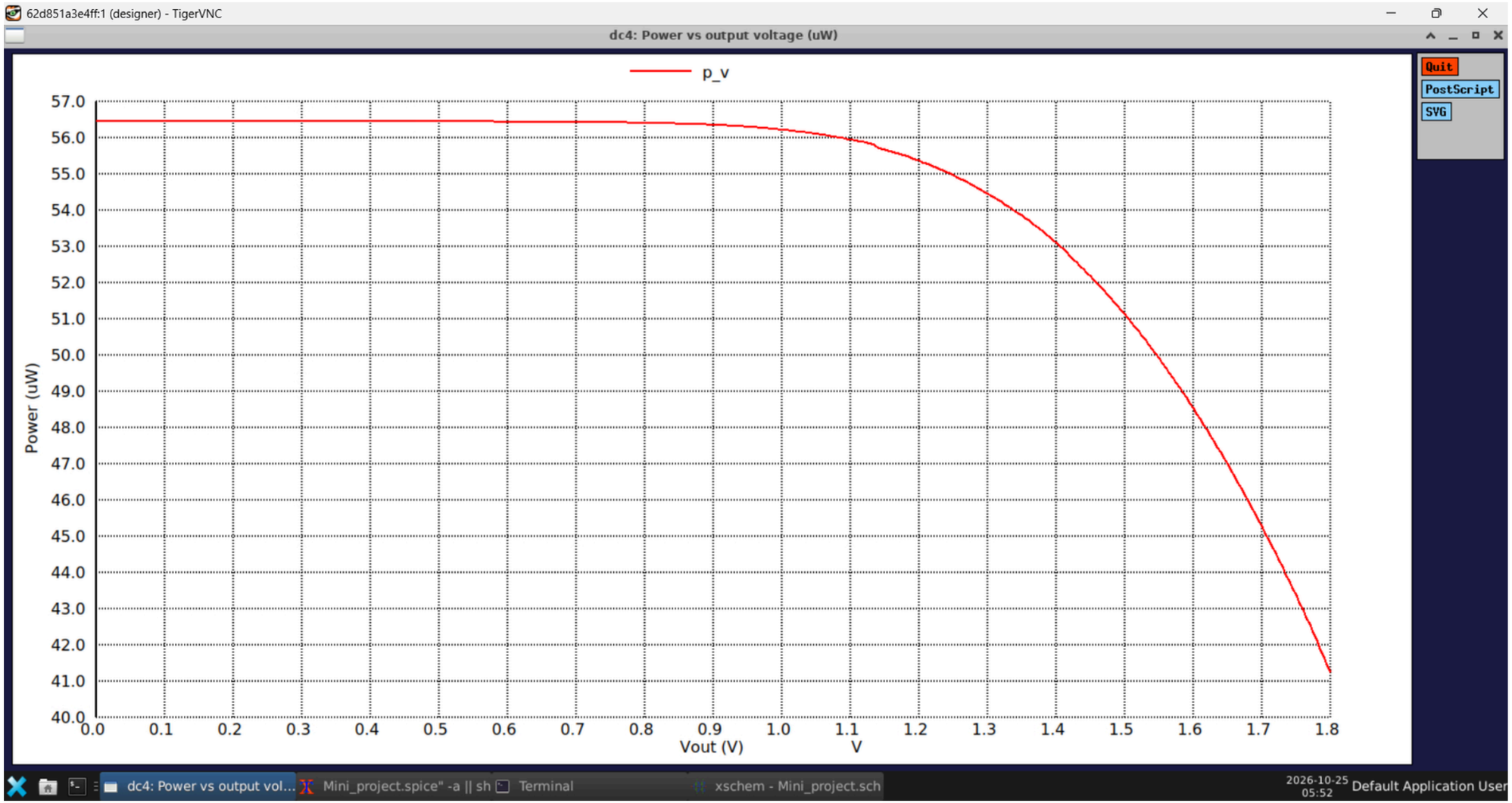
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Differential Phase



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Power Dissipation vs Vout



$$Ad = \boxed{38.084B} \qquad Ad = 20log_{10}(\frac{0.897917 - 0.290625}{6 \times 10^{-3}}) = \boxed{40.1dB}$$

$$Ad = gm3Rout \qquad Rout = Rup||Rdown$$

$$Rup = r3 = \frac{1}{gds} = \frac{1}{3.014 \times 10^{-7}} = 3.316M\Omega$$

$$Rdown = gm5r05r07 = 1.16 \times 10^{-4} \times \frac{1}{3.81 \times 10^{-7}} \times \frac{1}{1.10 \times 10^{-5}} = 27.678M\Omega$$

$$Rout = \boxed{2.961M\Omega}$$

$$Ad = gm3Rout = 26.4 \times 10^{-6} \times 2.961 \times 10^{6} = 78.176 = \boxed{37.861dB}$$

PERFORMANCE

Parameter	Results
Supply Voltage	**1.8V**
Differential Gain	**38.084dB**
Common mode Gain	**-23.222dB**
CMRR	**61.314dB**
-3dB Bandwidth	**51.37KHz**
Unity-gain Frequency	**4.11MHz**
Phase Margin	**77.31°**
Input Transconductance	**26.4µS**
V out	**0.697V**
Power Dissipation	**56.4µW**
Swing	**0.607V**
Remarks	**All in Saturation**

OBSERVATION

- At the nominal corner, all transistors operate in saturation. The differential gain is 38.08 dB, which is about 2 dB below the 40 dB target.
- The amplifier has a comfortable phase margin (about 77°) and a unity-gain frequency of about 4 MHz, so it is stable across the simulated conditions.
- The simulated output swing is 0.607 V is a small-signal output for a 6 mV differential input, so it understates the available range. The gap to the 1 V hand estimate comes mainly from the input pair's overdrive: the W/L chosen for 20 μS of gm gives Vov ≈ 0.69 V instead of the assumed 0.2 V. This pushes the tail (M1) to the edge of saturation and reduces the output range.
- The reference and output currents of the M4/M5 mirror match at every operating point, which shows the mirror tracks well.

PROCESS CORNER ANALYSIS

Corner	Tail Transistor	I ref	I out	I bias	V out	Total Current	Power
tt	20µ	10µ	10µ	11µ	0.697	31µ	56.4µ
ff	29µ	14.5µ	14.5µ	11µ	0.720	40.63µ	73.14µ
ss	7.45µ	3.725µ	3.725µ	10.9µ	0.643	18.41µ	33.14µ
fs	30µ	15µ	15µ	10.8µ	0.789	41.54µ	74.47µ
sf	4µ	2µ	2µ	11µ	0.559	15.73µ	28.32µ

PROCESS CORNER ANALYSIS

Corner	Differential Gain	Common Mode Gain	CMRR	-3dB Bandwidth	Unity Gain Frequency	V out Swing	Phase Margin
tt	38.08dB	-23.22dB	61.31dB	51.37KHz	4.1MHz	0.607V	77.31°
ff	31.1dB	-12.00dB	43.19dB	137.8KHz	4.9MHz	0.481V	73.69°
ss	46.83dB	-53.67dB	100.5dB	10.95KHz	2.4MHz	0.770V	79.28°
fs	25.00dB	-10.13dB	35.13dB	276.4KHz	4.7MHz	0.278V	74.66°
sf	51.44dB	-56.04dB	107.4dB	4.700KHz	1.7MHz	0.840V	78.93°

VOLTAGE CORNER ANALYSIS

Metric	VDD - 10%	VDD - 5%	VDD	VDD + 5%	VDD + 10%
Differential Gain	47.5dB	43.0dB	38.0dB	34.4dB	31.7dB
Common mode Gain	-56.4dB	-52.1dB	-23.2dB	-12.6dB	-9.7dB
CMRR	103.9dB	95.2dB	61.3dB	47.0dB	41.4dB
I tail	2.4µ	8.3µ	20µ	29µ	37.4µ
I ref	1.2µ	4.1µ	10µ	14µ	18.7µ
I out	1.2µ	4.1µ	10µ	14µ	18.7µ
I bias	9.4µ	10.3µ	11.1µ	12.2µ	13.1µ
V out	0.562V	0.630V	0.697V	0.744V	0.782V
Power Dissipation	21.3µW	33.6µW	56.4µW	75.5µW	91.1µW
Swing	0.719V	0.679V	0.60V	0.547V	0.485V
Remarks	All in deep Saturation	All in deep Saturation	All in Saturation	M1, M6 & M7 in Triode	M1, M6 & M7 in Triode

TEMPERATURE CORNER ANALYSIS

Metric	- 40°	0°	27°	85°	125°
Differential Gain	49.1dB	42.9dB	38.0dB	30.3dB	25.6dB
Common mode Gain	-56.6dB	-45.4dB	-23.2dB	-12.8dB	-11.5dB
CMRR	105.7dB	88.3dB	61.3dB	43.1dB	37.1dB
I tail	5.98µ	14µ	20µ	24µ	25µ
I ref	2.99µ	7µ	10µ	12µ	12.7µ
I out	2.99µ	7µ	10µ	12µ	12.7µ
I bias	11µ	11.3µ	11.1µ	11.2µ	11.2µ
V out	0.650V	0.678V	0.697V	0.703V	0.700V
Power Dissipation	31µW	45.6µW	56.4µW	64.4µW	66.0µW
Swing	0.85V	0.71V	0.60V	0.424V	0.272V
Remarks	All in deep Saturation	All in deep Saturation	All in Saturation	M1, M6 & M7 in Triode	M1, M6 & M7 in Triode

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FUTURE SCOPE

- Resize the input pair to cut overdrive. Increasing W of M2 and M3. This brings Vov from about 0.69 V toward 0.2 V, raises gm, and frees headroom at the tail. Rerunning the DC and AC sweeps to confirmation.
- Using a self-biased design for Vb. Replace the fixed 1.125 V source with a bias generator.
- Setting the input common-mode to about 0.97 V.
- Widen M1 to lower its Vdsat. Increase the tail's W/L so it stays in saturation with more margin.
- Making cascade pair to run in deep saturation varing the dimensions.
- Checking noise analysis, Off-set & slew rate analysis.

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RESOURCES/REFERENCES USED

- Behzad Razavi , Design Of Analog CMOS Integrated Circuits, Second Edition.
- Analog Integrated Circuit Design , David Johns , Ken Martin, University of Toronto.
- Phillip E. Allen Douglas R. Holberg crvlos Analog Circuit Design Second Edition

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Thank You!

Analog IC Design Mini-Project Presentation